

ABBA6 Implicit

Yoshida seven-stage composition of projected ABBA2 maps

1 Seven-stage composition

Let $\mathcal{P}_{q,t}$ be one complete symmetric projected ABBA2 step with signed duration q . The sixth-order model uses

$$\begin{aligned} a &= 0.78451361047755726382, & b &= 0.23557321335935813368, \\ c &= -1.17767998417887100695, & d &= 1.31518632068391121889, \end{aligned}$$

and $(w_1, \dots, w_7) = (a, b, c, d, c, b, a)$. Starting from $z^{(0)} = z_n$, it applies

$$t_j = t_n + h \sum_{\ell < j} w_\ell, \quad qquad z^{(j)} = \mathcal{P}_{w_j h, t_j}(z^{(j-1)}), \quad z_{n+1} = z^{(7)}.$$

The binary64 coefficients satisfy, up to round-off,

$$\sum_j w_j = 1, \quad \sum_j w_j^3 = 0, \quad \sum_j w_j^5 = 0.$$

Together with palindromy, these are the relevant odd-order cancellation conditions for a symmetric second-order base.

2 Projection at every factor

Every $\mathcal{P}_{w_j h, t_j}$ starts a new projection problem. In reduced form, with the usual diagonal operators $E, P, G, N = G^T$, factor j solves

$$M_j(\mu_j) = \Psi_{w_j h, t_j}(Ez^{(j-1)} + N\mu_j), \tag{1}$$

$$GM_j(\mu_j) + 2\mu_j = 0, \tag{2}$$

$$z^{(j)} = P(M_j(\mu_j) + N\mu_j). \tag{3}$$

Thus one outer step has seven projections and seven nonlinear solves. It is not a single projection around an unprojected seven-map base.

3 Configurations

Seven signed projected ABBA maps. Energy tracking is optional and passive. Each accepted projected factor contributes its signed shear-stage increment. There are seven nonlinear solves per complete step. The common state conventions are specified below.

4 Signed time path

Coefficients c in factors three and five are negative. Partial sums also leave the outer interval: the composition reaches approximately $t_n + 1.02009h$, then $t_n - 0.15759h$, later $t_n + 1.15759h$, and finally $t_n + h$. This non-monotone path is part of the real sixth-order composition; the dynamics must be defined at all internal times.

5 Order and finite-root qualifications

If the projected ABBA2 base is symmetric and second order, and each projection equation is solved to the relevant local root, the palindromic composition has designed global order six. Finite Newton or Broyden stopping, floating-point coefficient error, and non-smooth dynamics perturb this ideal statement. Diagnostics consequently retain work and residual data for every factor as well as outer-step aggregates.

6 Implementation correspondence

The public class is in `src/methods/extended/order6_implicit.py`; its Yoshida weights are in `extended/coefficients.py`. The common ABBA runtime repeats seven complete projected ABBA2 pairs. A single projection around the whole fourteen-stage composition would change the method and is not used.

The public names, constructor defaults and exports from `simulation` remain unchanged. Internal imports use the defining modules; old import adapters have been removed. The ordinary implicit path is `advance`, `solve_projection`, `compose`, then `direct_map` or `adjoint_map`. Arithmetic midpoint variants use one explicit average instead of a nonlinear projection. Adjoint stages evaluate at the start of their signed substep; direct stages evaluate at its end.

Newton and Broyden share the spatial equations and convergence services. Exact particle-block tangents, passive energy and optional observation use retained shear inputs; energy is evaluated only after accepting the spatial map, with the existing factor $1/2$ for physical κ . Time and momentum never enter the spatial root. The new internal family name does not enable the retired full time/momentum projection.

Current module responsibilities and diagrams are documented in `docs/models/extended/simulation/extended-simulation-architecture.md`.

7 Four state formulations and passive energy tracking

For one planar particle the supported internal representations are

physical	$z = (x, y)$	$\mathbb{R}^2$,
physical with energy	(z, t, κ)	$\mathbb{R}^4$,
duplicated	$(u, v), \quad u, v \in \mathbb{R}^2$	$\mathbb{R}^4$,
duplicated with energy	(u, v, t, κ)	$\mathbb{R}^6$.

Classical methods use the physical rows; ABBA and BM4 use the duplicated rows. `track_energy` selects the energy variant. For N planar particles the respective dimensions are $2N$, $4N$, $4N$, and $6N$: one time entry and one momentum per particle. All time entries equal the integration time. Classical full-cyclotron runs instead use $4N$ or $6N$ entries. Accepted spatial copies coincide; only one physical copy is exported. Every coordinate block contains N values in component-major order. The formulation validates particle times and aligns output round-off; the integrator does not interpret coordinate indices.

The normalized passive balance is

$$\dot{\kappa} = -\partial_t H(t, z), \quad \kappa(t_0) = 0, \quad \Delta K(t) = H(t, z(t)) + \kappa(t) - H(t_0, z_0).$$

Splitting accumulators from both copies are divided by two. The physical Hamiltonian may vary with time; ΔK measures its computed balance. Tracking leaves physical trajectories, nonlinear stopping rules and adaptive acceptance unchanged. Dynamics must supply the momentum derivative, including zero for an autonomous Hamiltonian.

Hairer's constraint is solely $u - v = 0$: the reduced multiplier has $2N$ entries and ABBA's simultaneous spatial solve has $6N$ unknowns. Neither t nor κ is projected. The former eight-component projection is retired; `state_extension='fully_extended'` raises migration guidance. Dynamics, formulation, method and integration remain separate. The shared state classes own layouts, diagonal embedding and history extraction in

```
src/formulations/state.py
```

Each integration initializes a fresh method instance. Observers receive physical states; energy histories are in `Solution.diagnostics`. The `extended_time` history has shape (N, n_{samples}) . Fixed-step shadow samples leave main states and counters unchanged. Adaptive energy quadrature uses accepted dense output; its backend state and Jacobian remain physical.